

Reviewing TG158

Measurement and calculation of doses outside the treated volume from
EBRT



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1 Introduction

What is outfield dosimetry

- Improved conformality has not mitigated the problem of doses to normal tissues outside the treated volume.
- With modern treatment planning systems, high-dose regions and areas within the primary-beam path are typically well described. However, the accuracy of dose determinations beyond a few centimeters outside the treatment field edge is usually poor.
- The dose outside the PTV is referred to as “nontarget dose.”
- Nontarget dose is delivered by nontarget radiation and can be subdivided into two categories

Term	Definition
In-field	within a primary field border , such as entrance and exit dose along the beam path.
Out-of-field	not only outside of the PTV but also outside of any primary field edge — dose deposited by stray, or secondary, radiation.

What is outfield dosimetry

Nontarget dose classification

- Nontarget dose can be classified as one of three approximate dose levels. [?]

Dose	Definition
High doses	(> 30 Gy or $> 50\%$ of the prescription dose) are typically directly optimized during the planning process.
Intermediate doses	($3 - 30$ Gy or $5-50\%$ of the prescription dose) are also often addressed during optimization but are not generally the focus of the treatment plan
Low doses	(< 3 Gy or $< 5\%$ of the prescription dose) are typically not considered during optimization and are often poorly modeled by the TPS.

References I



Rebecca M Howell, Sarah B Scarboro, S F Kry, and Derek Z Yaldo.
Accuracy of out-of-field dose calculations by a commercial treatment
planning system.
Phys. Med. Biol., 55(23):6999, November 2010.